SWAPNIL MITRA

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Software Engineer with 3+ years of experience in system architecture, enterprise middleware integration, and AI-enabled tooling across .NET Core, Azure, RabbitMQ, React, and cloud-native distributed systems. Seeking engineering roles building scalable, high-availability production services.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C#, Python, Rust, SQL, PostgreSQL, Solidity
Frameworks: .NET Core, ASP.NET, React, Next.js, Flutter, Express, Fluent UI, MSAL, LangChain, Streamlit, Tailwind CSS, Wagmi
Cloud & Middleware: Azure Functions, API Management, Logic Apps, Azure DevOps, GitHub Actions, RabbitMQ, Docker, AWS (S3, Bedrock, Transcribe, Polly)
Tools & Platforms: Prometheus, Grafana, Microsoft Graph API, SharePoint, SQL Server, Git, Hardhat, Prisma, Drizzle, pgvector, Pinata (IPFS), Vitest

PROFESSIONAL EXPERIENCE

Adrosonic | **Research and Innovation Engineer** · *Jul 2023 – Present*

- Technical Lead for a team of 4 on an enterprise broker management platform, owning solution architecture across four delivered Azure workstreams: SSO for the broker-agency user base, then quote-bind, cancellation, and amendment insurance flows.
- Architected the middleware integration layer feeding downstream quote-aggregator services in Australia: a JSON transformation engine on Azure Functions, API Management gateways with OAuth 2.0 and Managed Identity policies, and Azure Application Analytics telemetry, cutting integration development from ∼6 weeks to ∼2 at a 99.9% payload delivery rate.
- Re-engineered an email filing agent and SharePoint DMS pipeline onto .NET Core and RabbitMQ pub/sub with deduplication, dead-letter queuing, and mirrored cluster queues in Docker, plus Prometheus dashboards, for a ∼40% drop in average message latency.
- Migrated a legacy COM-based Outlook add-in to a React and Fluent UI web add-in with MSAL SSO, Microsoft Graph integration, and IndexedDB offline attachment caching; deployed to 10,000+ users.

PROJECTS

MemVault | **github.com/swap-mitra/memvault** · *Aug 2026*
Local-first, hash-chained memory engine for AI agents with fully explainable retrieval.

- Built the engine in Rust with an append-only, hash-chained ledger of every fact and retrieval; cryptographic erase destroys the content key while the chain still verifies end to end.
- Fused usearch ANN and BM25 ranks by reciprocal rank fusion with decay weighting and token-budget cutoffs, with per-candidate provenance that replays any past retrieval; shipped as a CLI, an MCP stdio server, and PyO3 bindings.

Groundwork | **github.com/swap-mitra/groundwork** · *2024 – 2026*
Evidence-grounded outbound email drafting gated behind explicit human approval.

- Built a multi-tenant outreach workspace (Next.js, Drizzle, Postgres + pgvector) that ingests a job URL behind an SSRF guard, then drafts evidence-cited email using hybrid retrieval over a halfvec HNSW index.
- Enforced human-in-the-loop delivery with append-only draft versions, reviewer-attributed approvals, and an audit trail; ran ingestion and generation as Inngest workflows with bounded retries and durable rate limits.

City Vault | **github.com/swap-mitra/city-vault** · *2025 – 2026*
Multi-tenant records governance with 6-role RBAC and an IPFS audit trail.

- Architected the platform on Next.js 15, Prisma, and Neon Postgres with 6-role RBAC, full document lifecycle management, IPFS pinning via Pinata, and append-only audit trails.
- Added a sliding-window rate limiter with per-key tracking and a Vitest suite over route handlers and governance workflows.

AGRISETU | **github.com/hackerslash/agrisetu** · *Feb – Apr 2026*
Voice-first agricultural marketplace for collective group ordering.

- Engineered the marketplace (Flutter, Next.js, Express, Turborepo) processing PCM audio through AWS Transcribe in 11 languages, GLM-4.7 Flash on Bedrock, and Polly TTS.
- Deployed a RAG pipeline on Titan Text Embeddings v2 and a Bedrock Knowledge Base, eliminating hallucinated answers on dialect queries.

EDUCATION

Indian Institute of Technology, Kharagpur | **B.Tech, Biotechnology and Biochemical Engineering** · *May 2023*

EXTRA-CURRICULARS

- Associate Manager, E-Cell IIT KGP: organised Global Entrepreneurship Summit 2021, LSM 2020, EAD 2020, Empresario 2020.
- National Service Scheme Volunteer, Silver Medal (2019); Club Service Director, Rotaract Club of Calcutta North East.